

$$X\beta = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \xrightarrow[\text{column} \rightarrow \text{row}]{\text{transpose}} (X\beta)^{\top} = \begin{bmatrix} 3 & 4 \end{bmatrix}$$

$$\beta^{\top} X^{\top} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1+2 & 1+3 \end{bmatrix} = \begin{bmatrix} 3 & 4 \end{bmatrix}$$

same row: the transpose of a product is the product of the transposes, in reverse order